COSMOLOGY

Problem Set 6: Relativistic Perturbation Theory

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These are the end of chapter problems for Chapter 6 of the notes. Equation numbers refer to the full notes.

1. Tensors Are Gauge Invariant

Consider a transverse-traceless perturbation h_{ij} of the spatial metric.

a). Under a gauge transformation with $\xi^\mu = (T, \xi^i)$, show that δg_{ij} shifts by $-a^2(\partial_i \xi_j + \partial_j \xi_i + 2\mathcal{H} T \delta_{ij})$. Decompose $\xi^i = \partial^i L + L^i_\perp$ with $\partial_i L^i_\perp = 0$.

b). Show that neither T, L nor $L^i_\perp$ can generate a transverse traceless contribution. Conclude that h_{ij} is gauge-invariant on its own, without needing to form any combination.

2. Gravitational Waves In The Matter Era

a). Show that during matter domination ($\mathcal{H} = 2/\eta$), the tensor equation $h_k'' + 2\mathcal{H} h_k' + k^2 h_k = 0$ has the regular solution $h_k = 3 h_k^{\rm prim}\, j_1(k\eta)/(k\eta)$, where $j_1(x) = (\sin x - x \cos x)/x^2$.

b). At what frequency today would we observe a gravitational wave that re-entered the Hubble radius at matter-radiation equality? (Use $k_{\rm eq} \approx 0.01$ Mpc^{-1} and $f = ck/2\pi$.) Why do CMB B-mode experiments probe such long wavelengths, while LIGO probes very short ones?

3. Synchronous Gauge

Instead of Newtonian gauge, one can choose $A = B = 0$, so that the coordinates are attached to freely falling observers (this is what the CMB code CAMB uses). Using (6.13), show that this does not fix the gauge completely: there is a residual freedom $T = C(\boldsymbol{x})/a$ with C an arbitrary function of space. What does this residual freedom correspond to physically?

4. The Integrated Sachs-Wolfe Effect

A CMB photon travelling through a time-dependent potential picks up an extra temperature shift $\Delta\Theta = \int (\Phi' + \Psi') d\eta$ along its path. Explain why this vanishes during matter domination, and why it does not vanish at late times when dark energy dominates (use the growth

equation of the last chapter and Poisson's equation, $\Phi \propto D_+(a)/a$ on subhorizon scales). This late-time effect is one of the ways the CMB detects dark energy.